DATE

09/10/2024

ACOUSTIC ENERGY HARVESTER

MAD 2024

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Alina hang
Kevin Ni

BACKGROUND AND AIM

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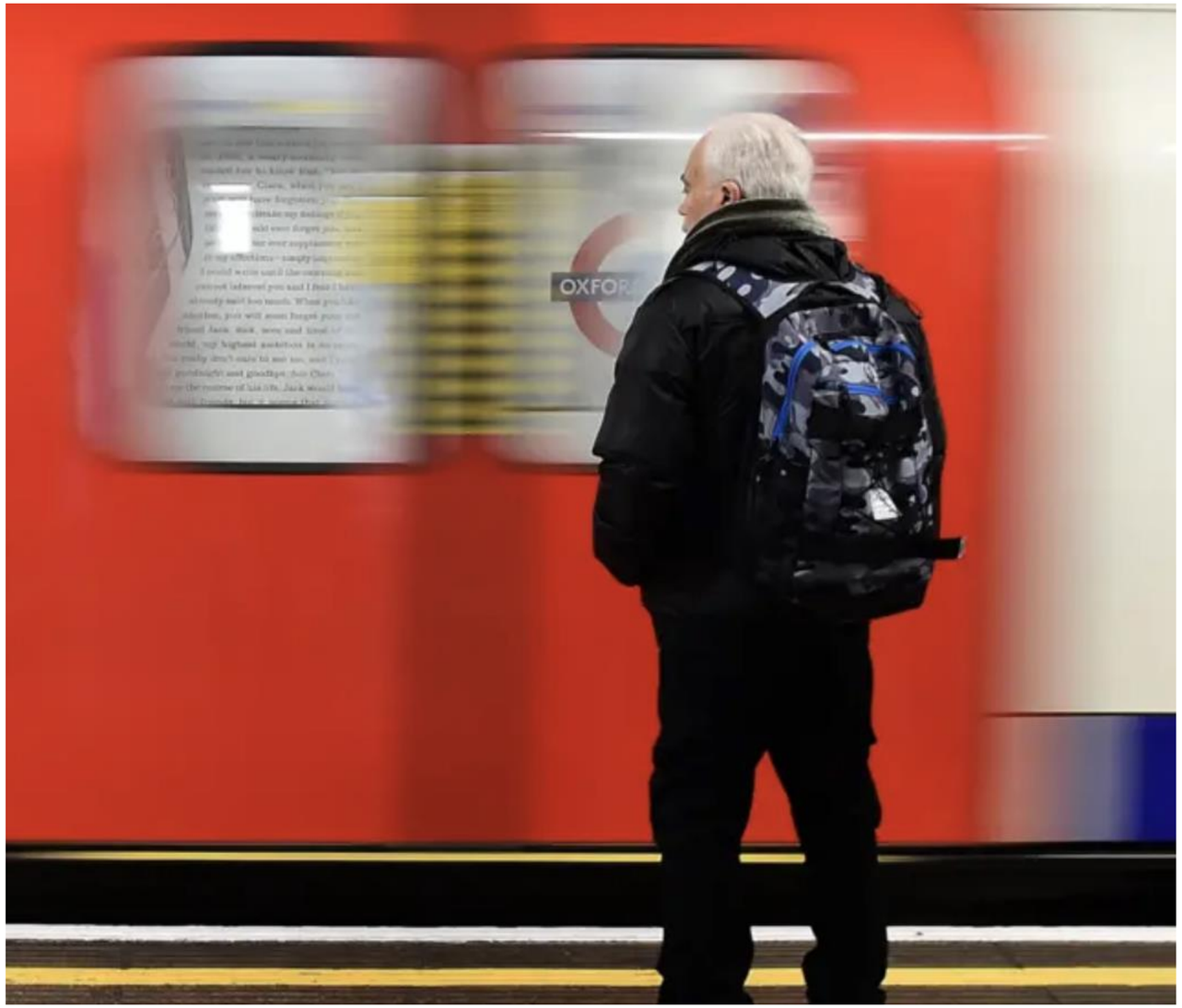
Background

- Energy scarcity and noise pollution
- Urban noise pose health risks (>120dB)
- Noise is an underutilized resource

Aim

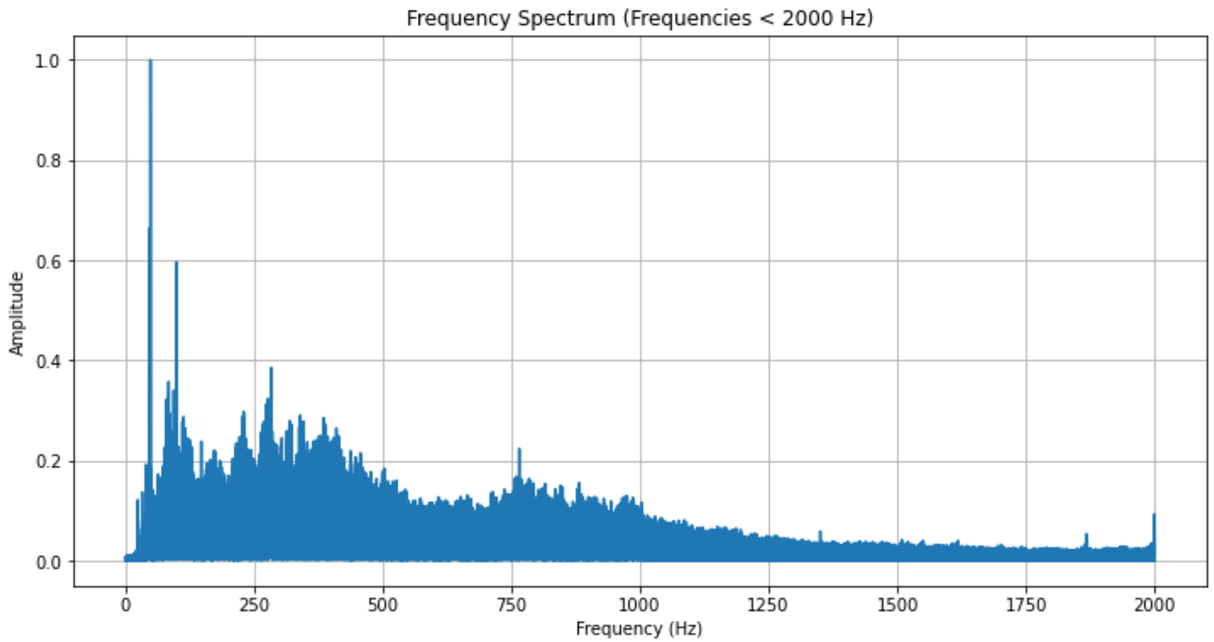
Develop an acoustic metamaterial to absorb subway noise and convert it into electrical energy.

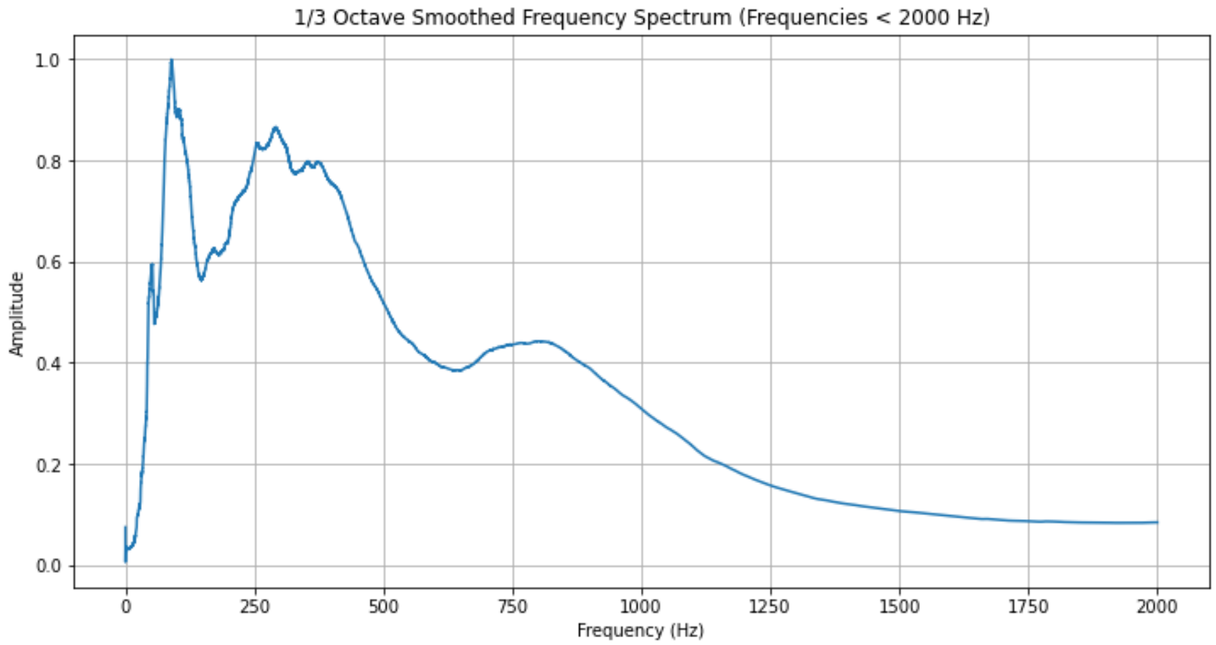


DESIGN AND PERFORMANCE

NOISE SPECTRUM

- Victoria Line
- 1/3 octave smoothing
- three peaks: 90 Hz, 315 Hz, and 850Hz





ACOUSTIC METAMATERIAL

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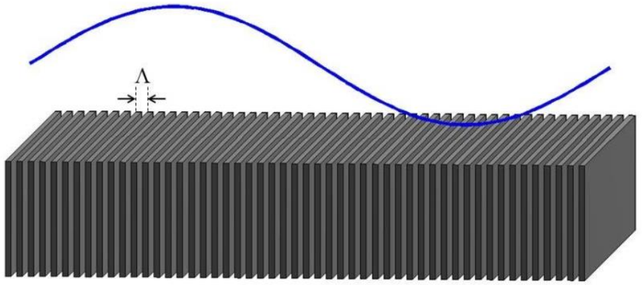
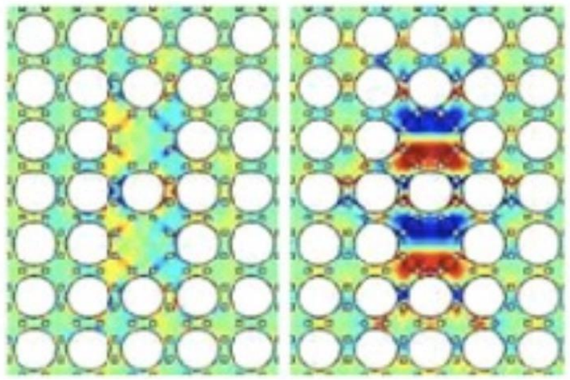
Local Resonance → Unusual Properties

Sub-Wavelength
Sound Manipulation

$\Lambda << \lambda$



Defect Mode



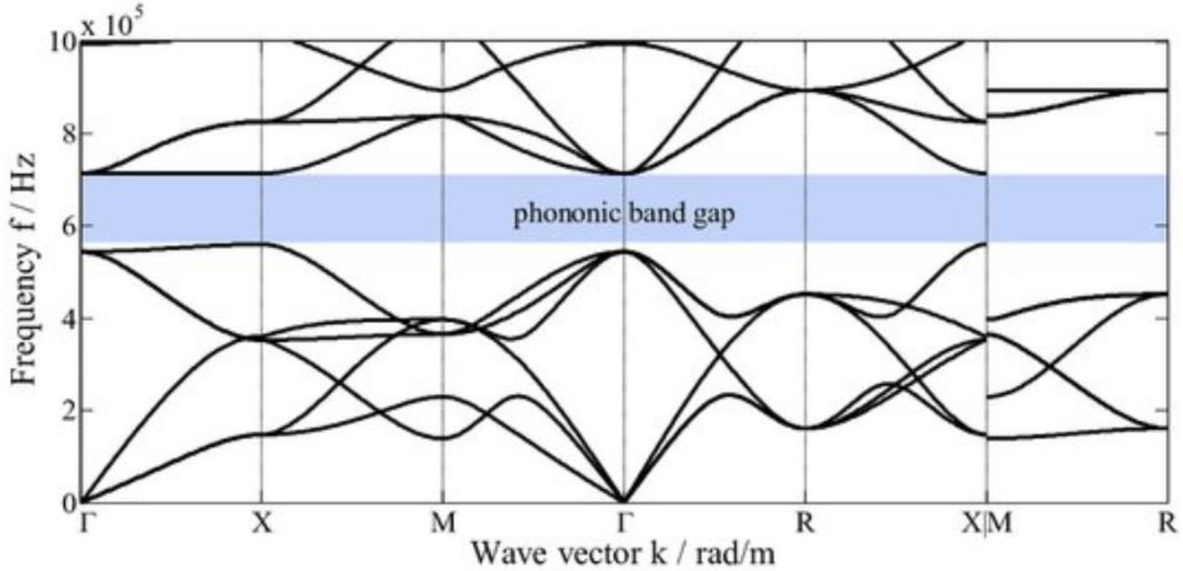
Perfect Low Frequency Absorption

Increased energy harvesting efficiency

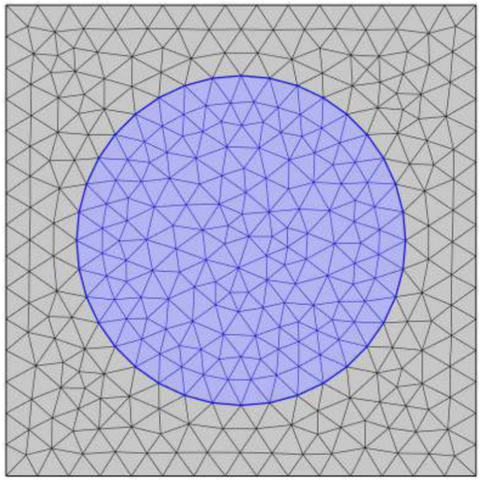
Band Gap

BAND GAP

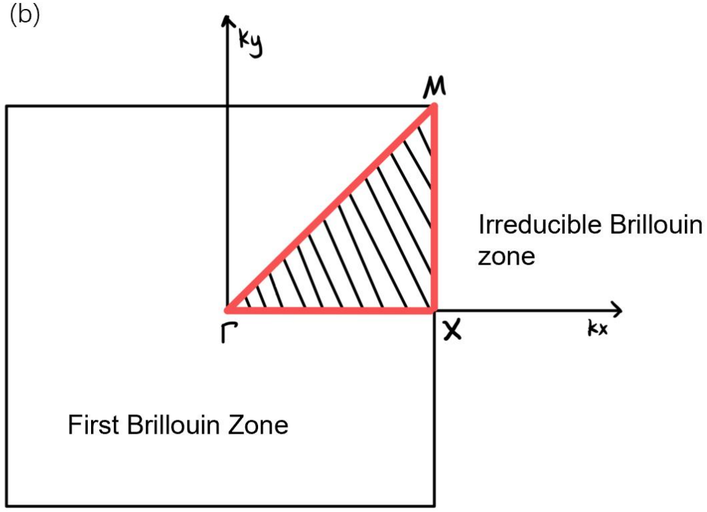
- Finite Element Method (FEM) in COMSOL
- Bloch's Boundary Conditions – wave propagation
- Irreducible Brillouin Zone -- symmetry



(a)



(b)



- Eigenvalue problem about stiffness matrix (elasticity modulus) and mass matrix (density and Poisson's ratio)
- Band gap exists when eigenvalue absent in all directions

DL-DATA SET

- **Material Parameter**

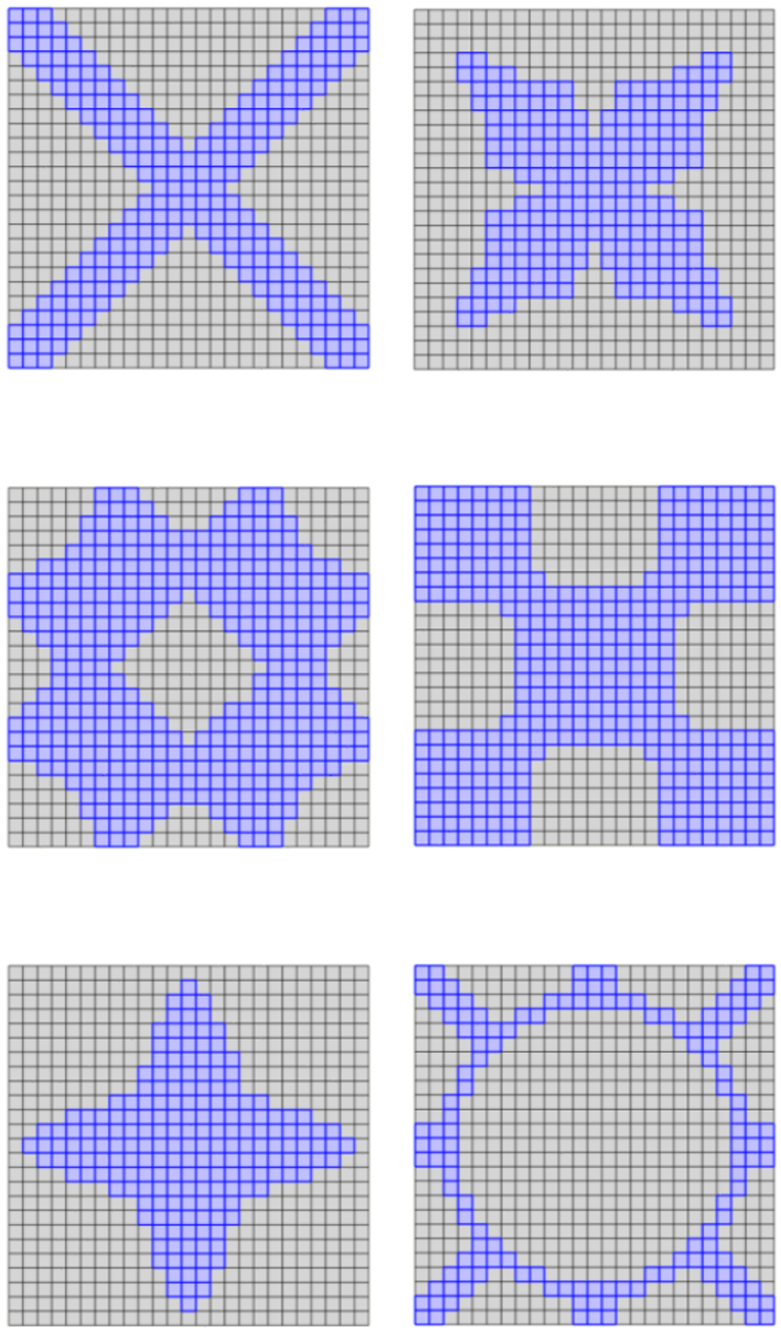
NumPy.Random function:

1. *Poisson's ratio (0.45 to 0.499)*
2. *Elasticity modulus(1 to 20 MPa)*
3. *Density (1.0 and 1.4 g/m³)*

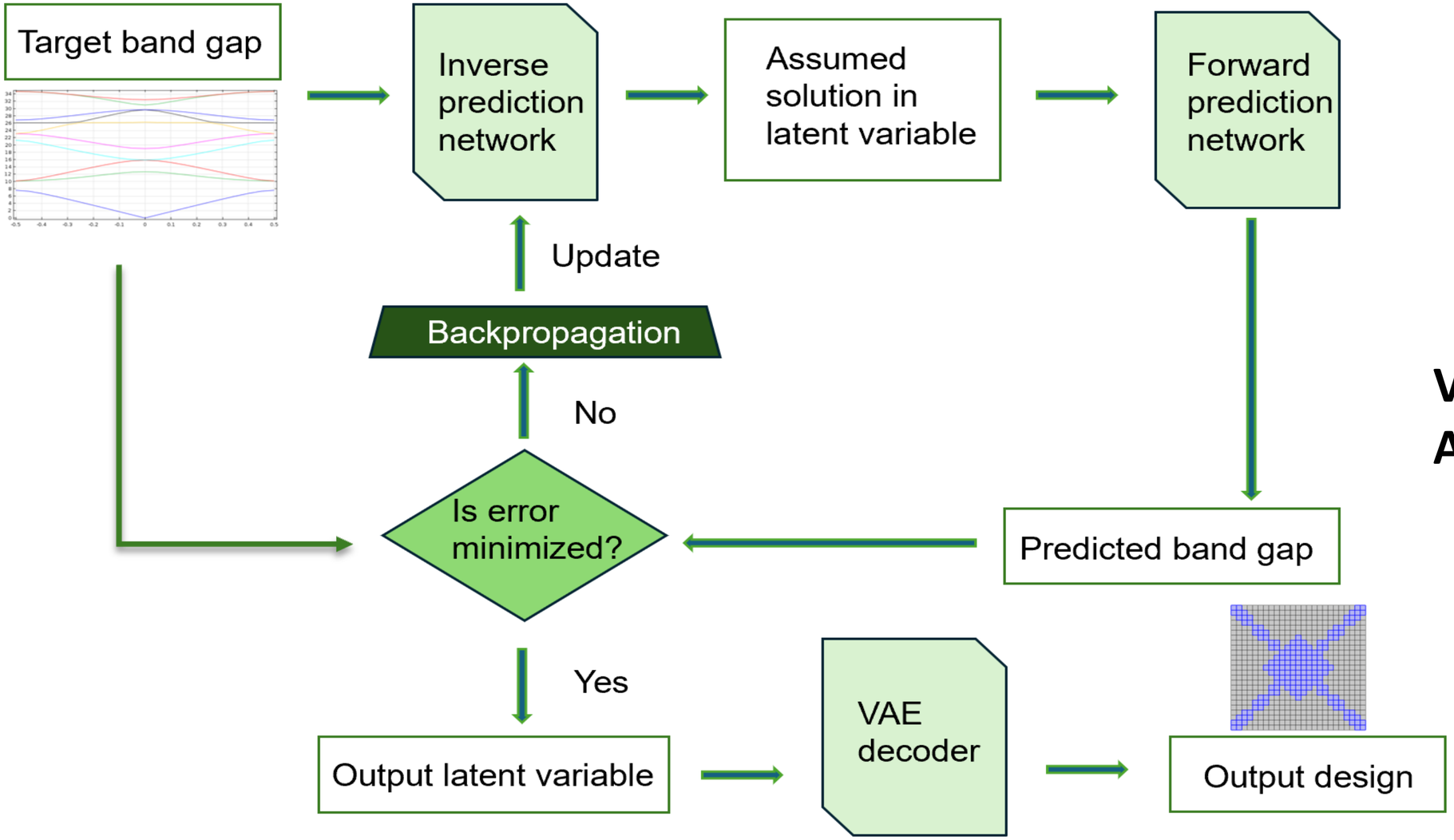
- **Geometry Parameter**

Topological Configurations

1. *25 × 25 grid*
2. *Filling ratio between 0.3 and 0.65*
3. *Scatterers form single connected domain*



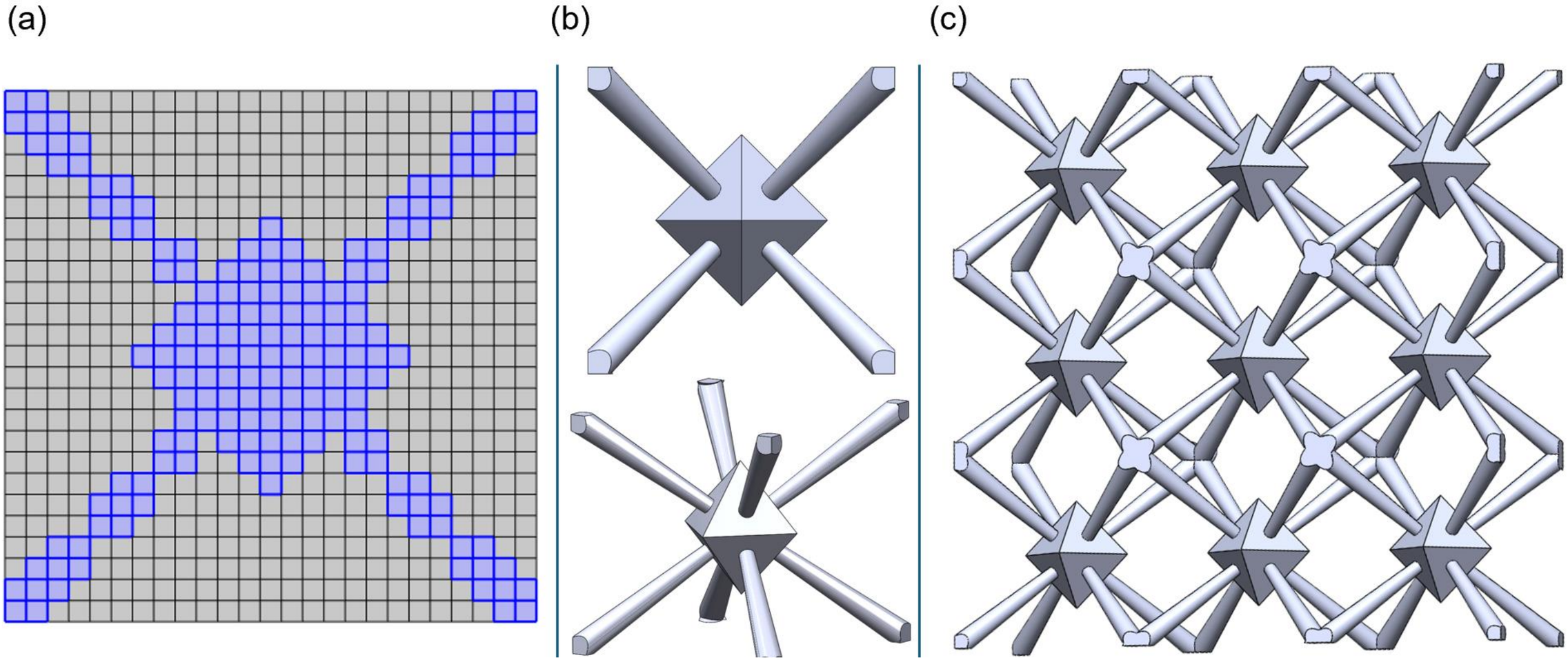
DL-INVERSE PREDICTION



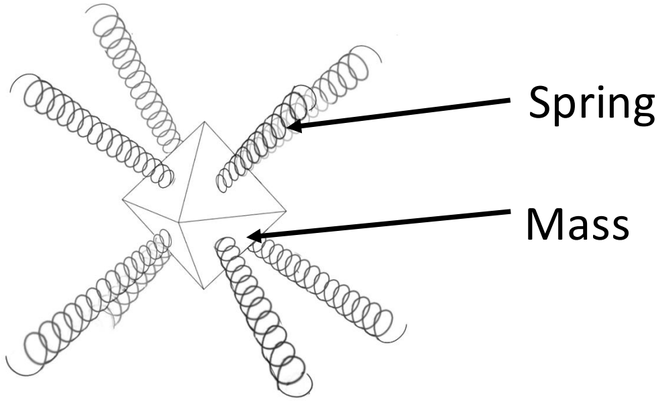
Variational Autoencoder (VAE)

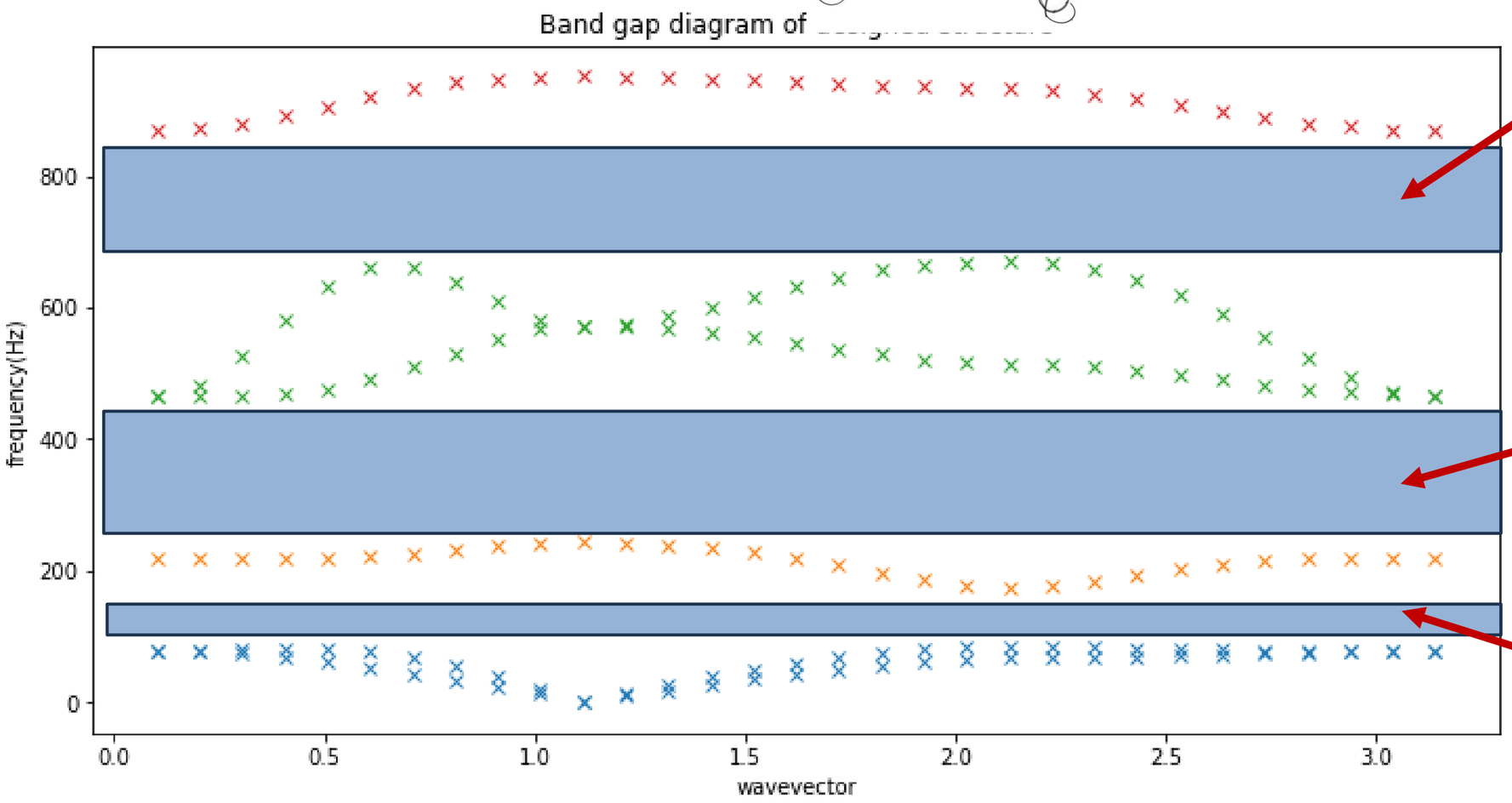
DL-RESULTS

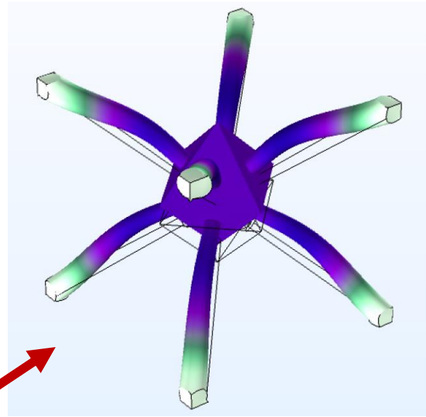
- Elasticity Modulus: 10 MPa
- Poisson's Ratio : 0.48
- Density:1.2 g/m³

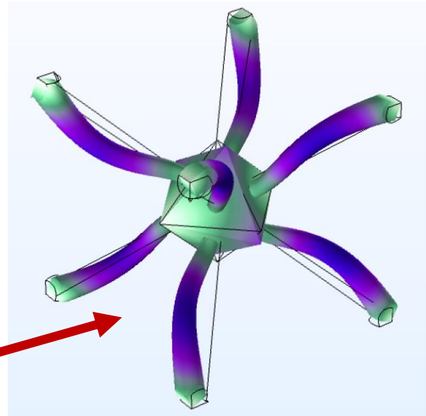


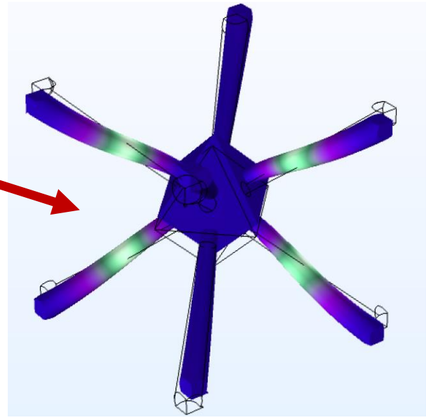
DL-RESULTS


Spring
Mass


Band gap diagram of
frequency(Hz)
0
200
400
600
800
0.0
0.5
1.0
1.5
2.0
2.5
3.0
wavevector

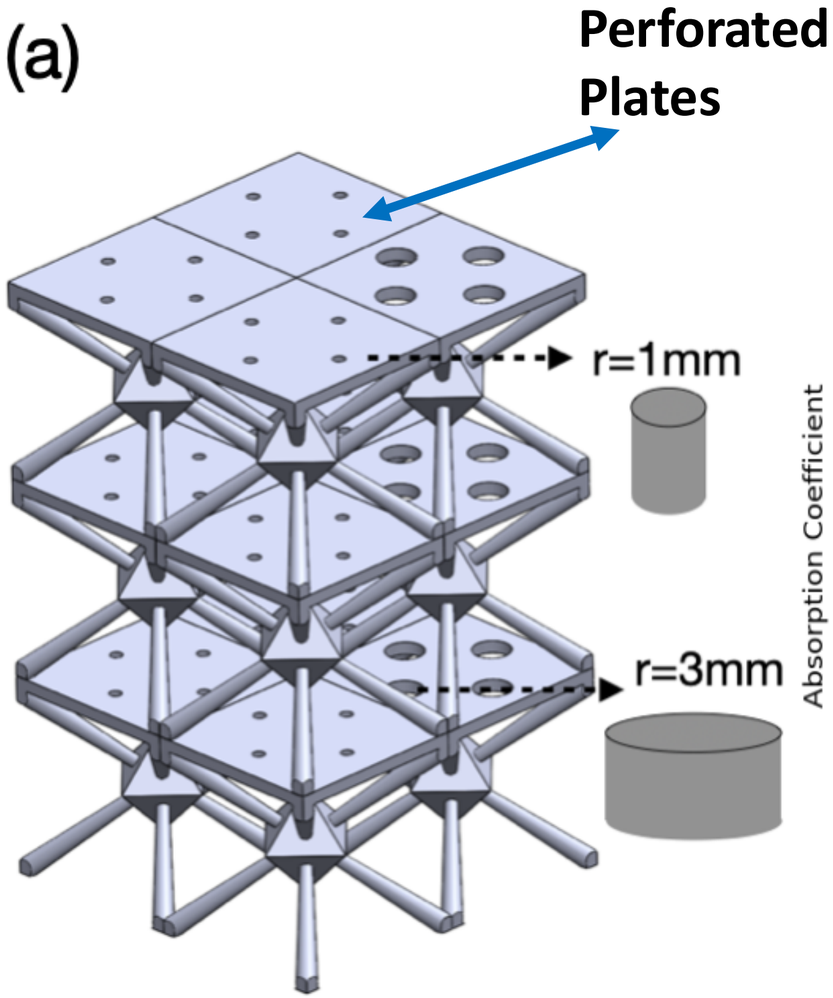


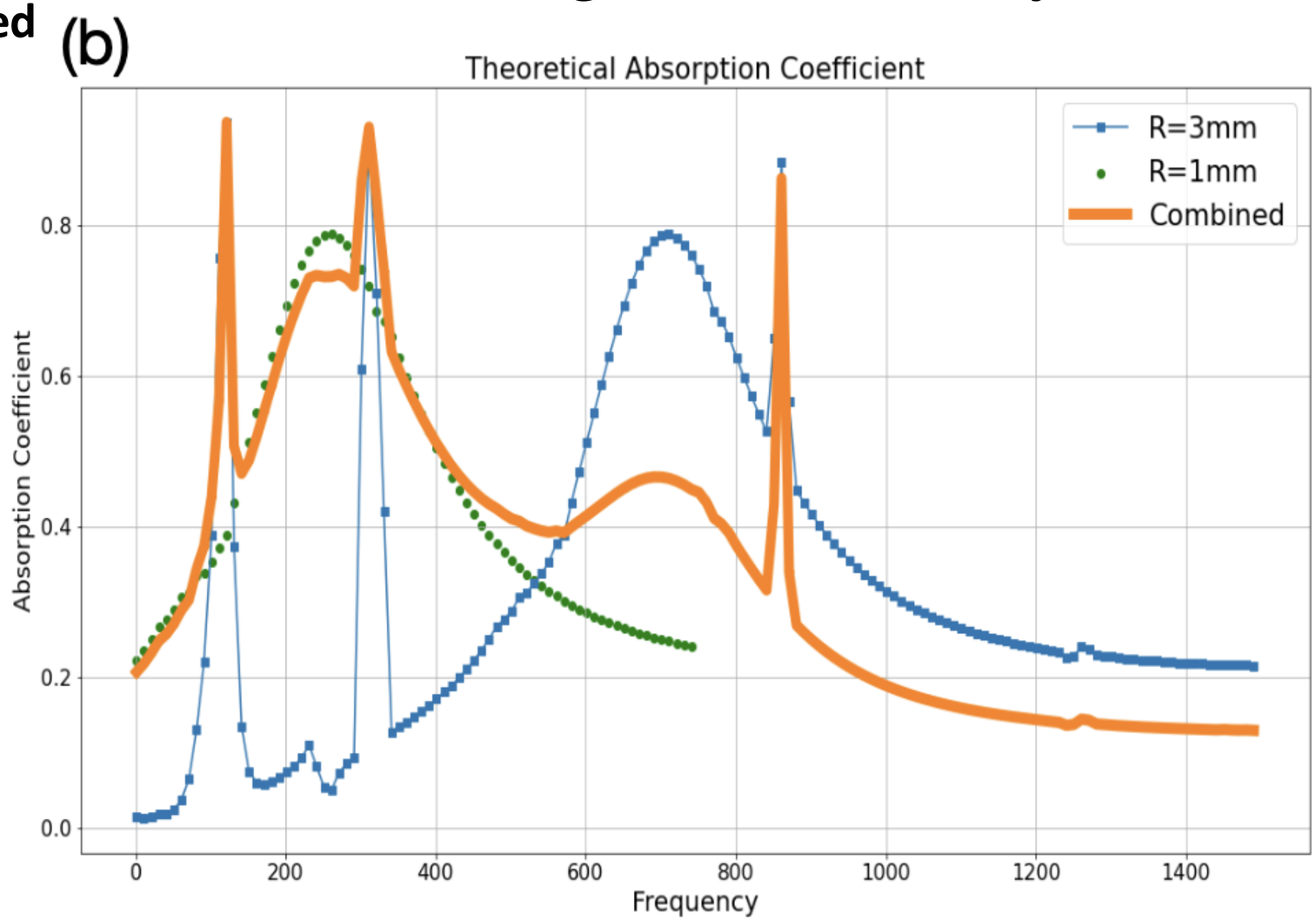




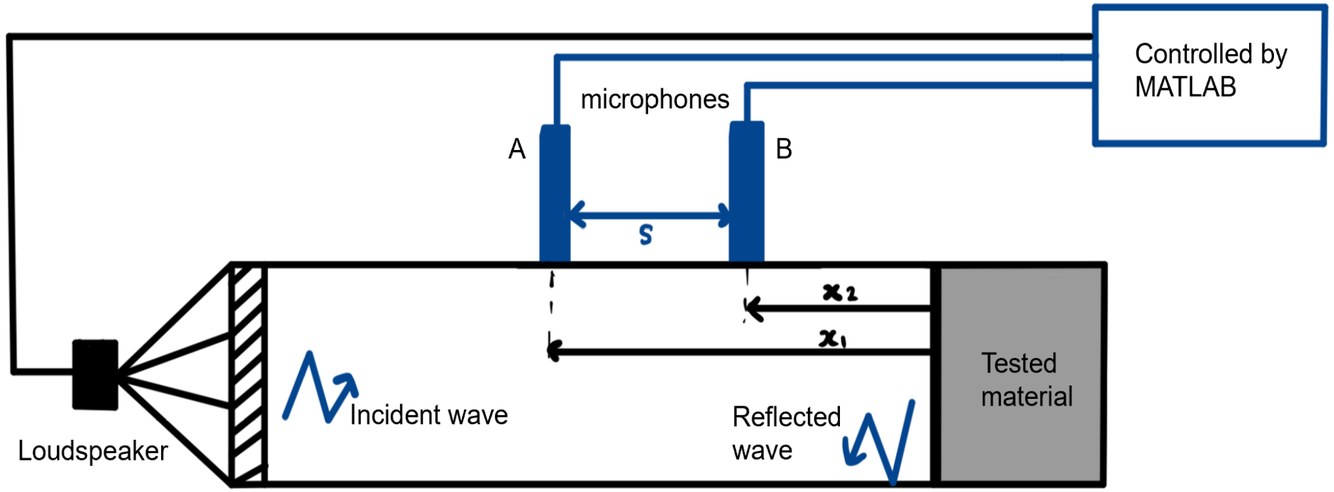
ABSORPTION

Cascading Resonator Theory

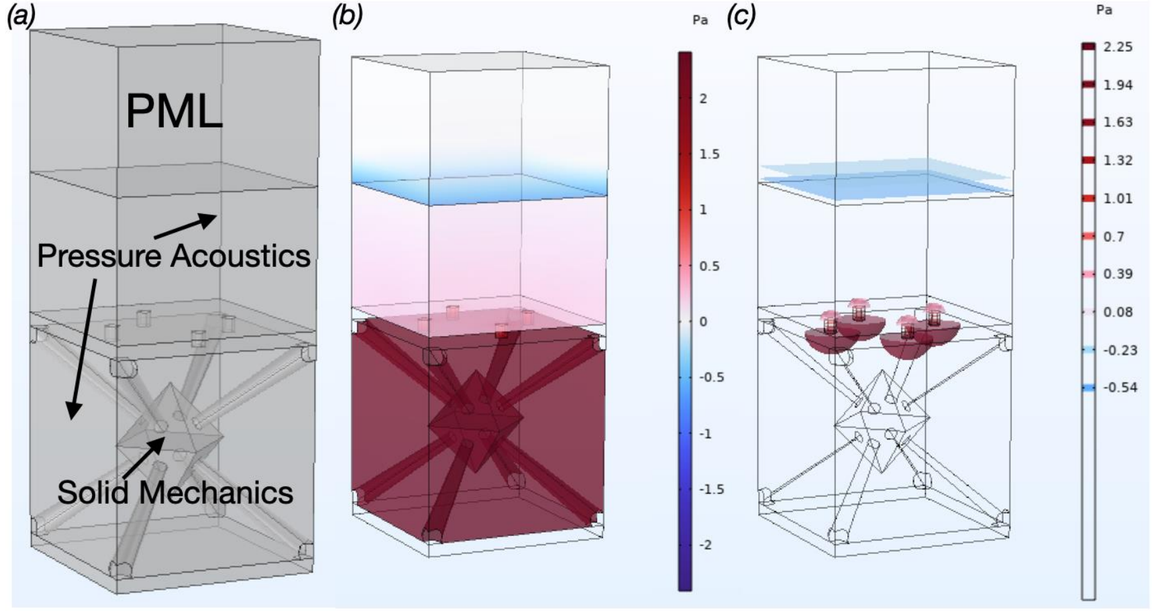




ABSORPTION

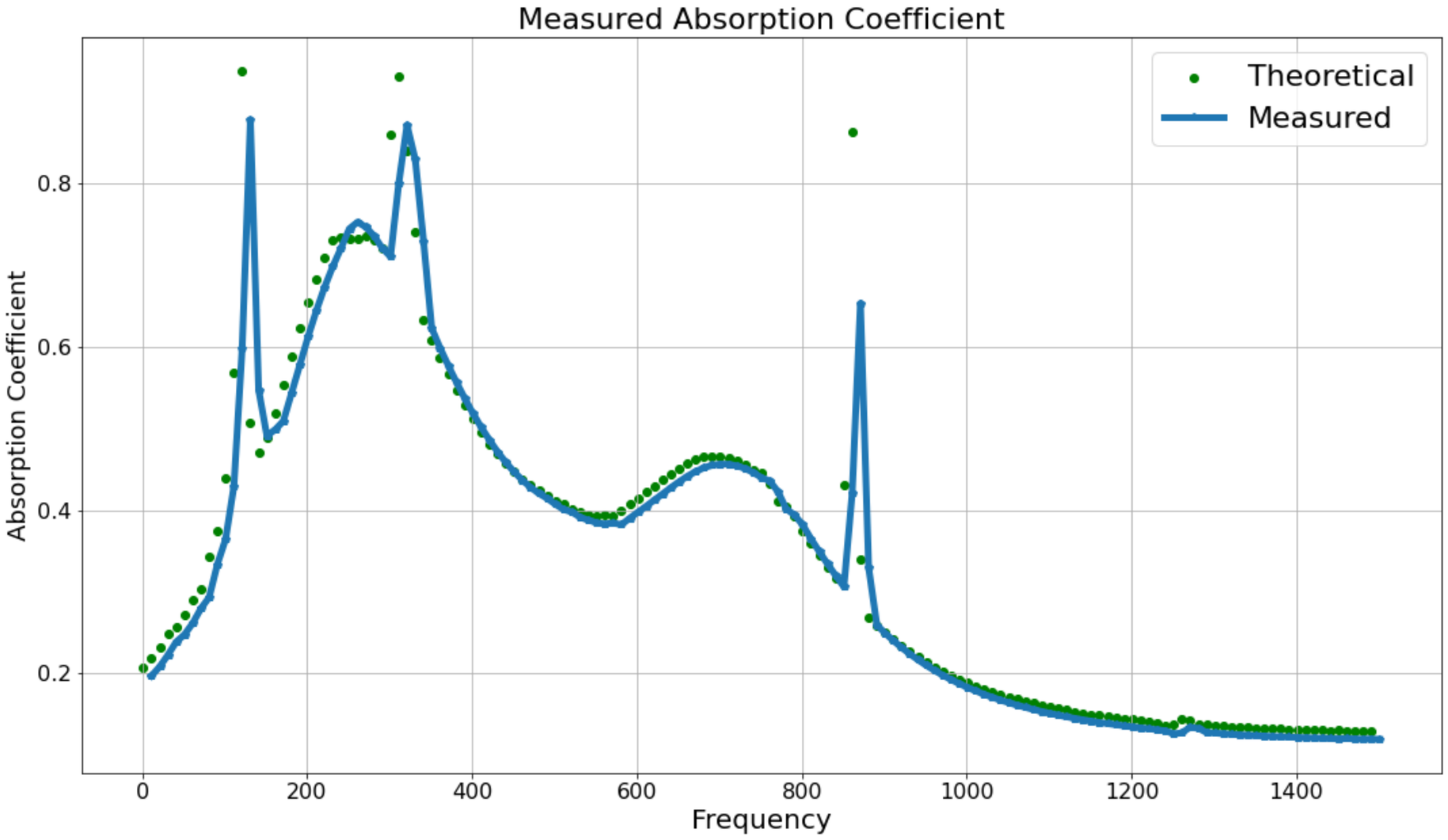


- **Simulated by COMSOL**



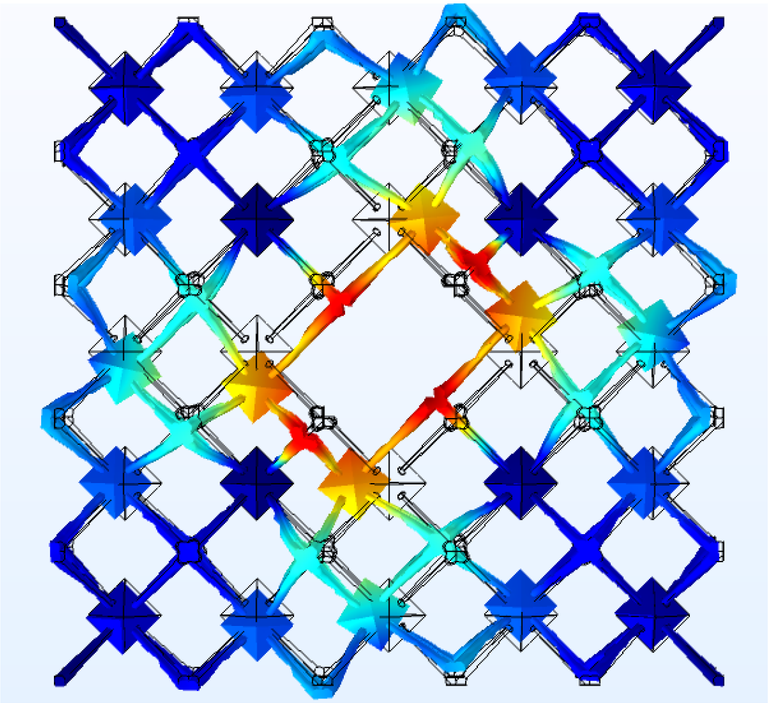
- **Measured by Impedance Tube**

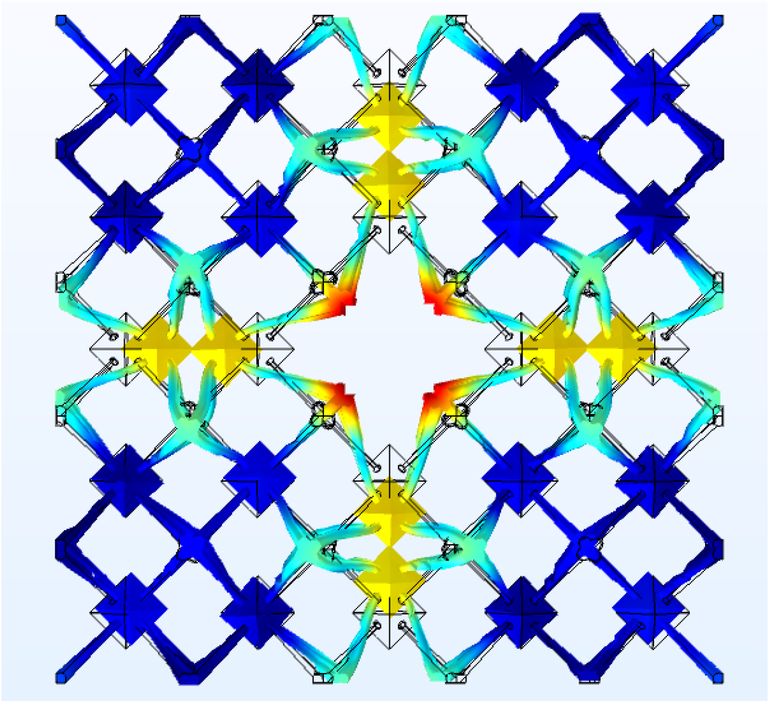
ABSORPTION


Measured Absorption Coefficient
Theoretical
Measured
Absorption Coefficient
0.2
0.4
0.6
0.8
0
200
400
600
800
1000
1200
1400
Frequency

ENERGY LOCALIZE

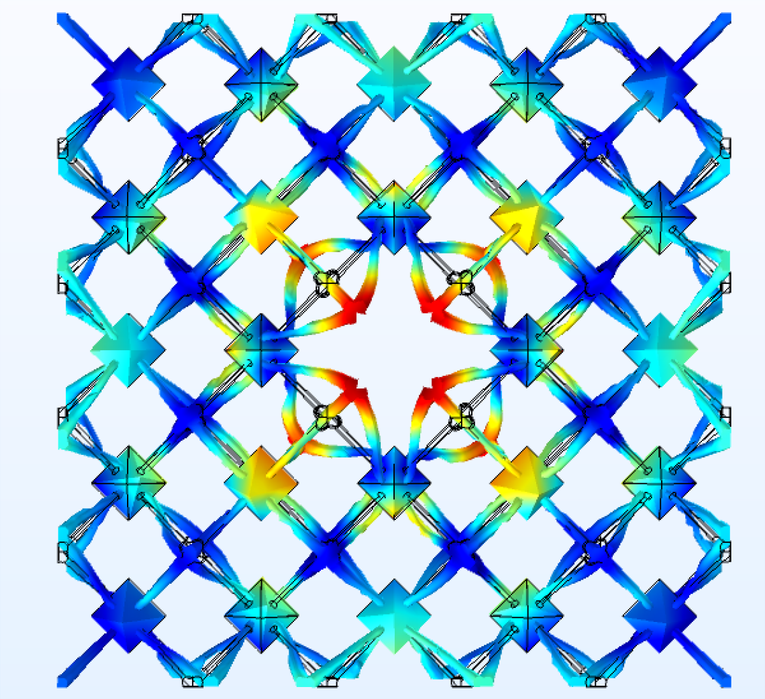
f = 90.12Hz



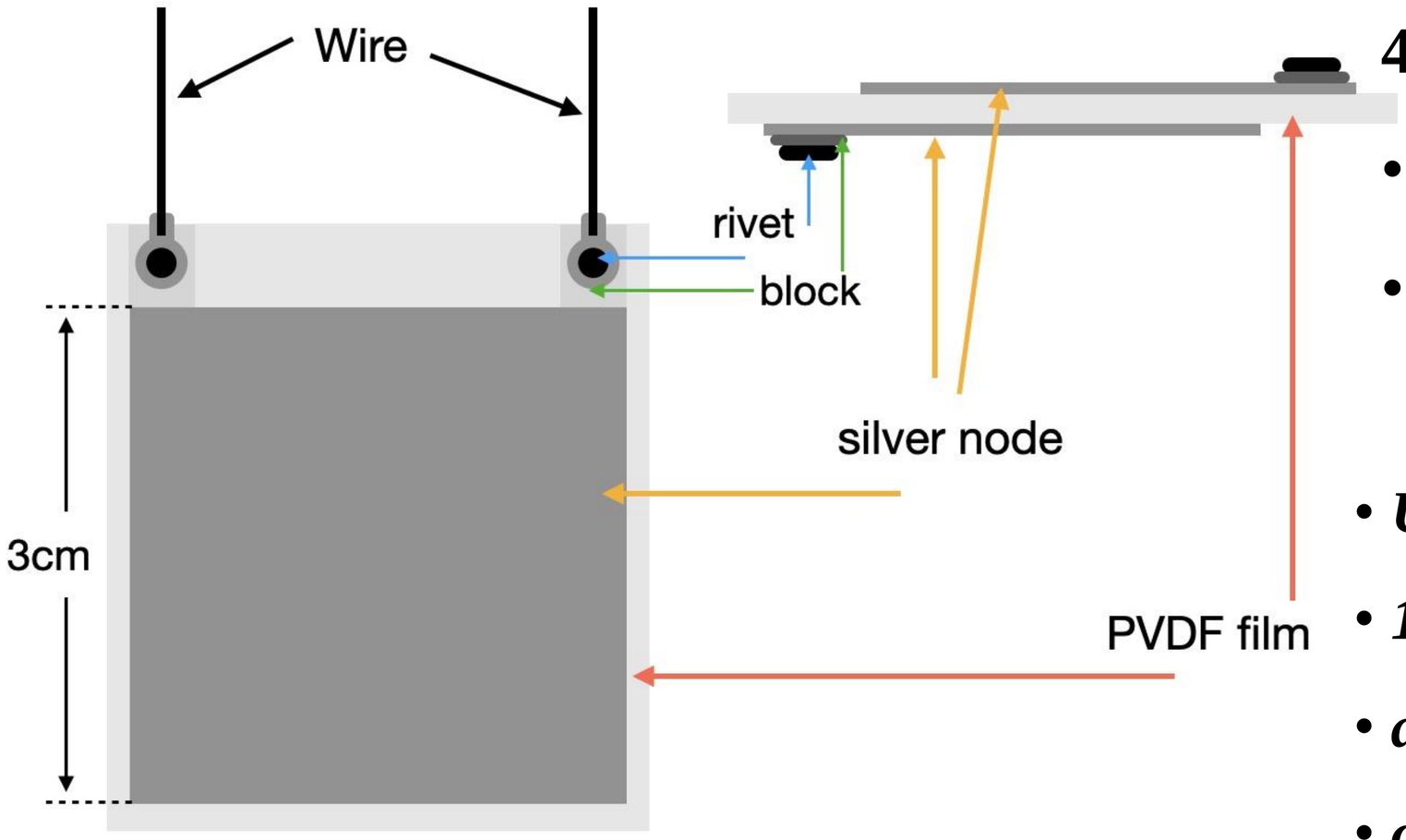


f = 351.7Hz

f = 855.36Hz



ENERGY TRANSDUCE



45um PVDF Pizo Film

- High piezoelectric coefficients
- Low-freq impedance match

- ***Under 100dB***
- ***1 kΩ resistor***
- ***average voltage: 2.37±0.29V.***
- ***output power: 5.62mW±1.37μW***

04

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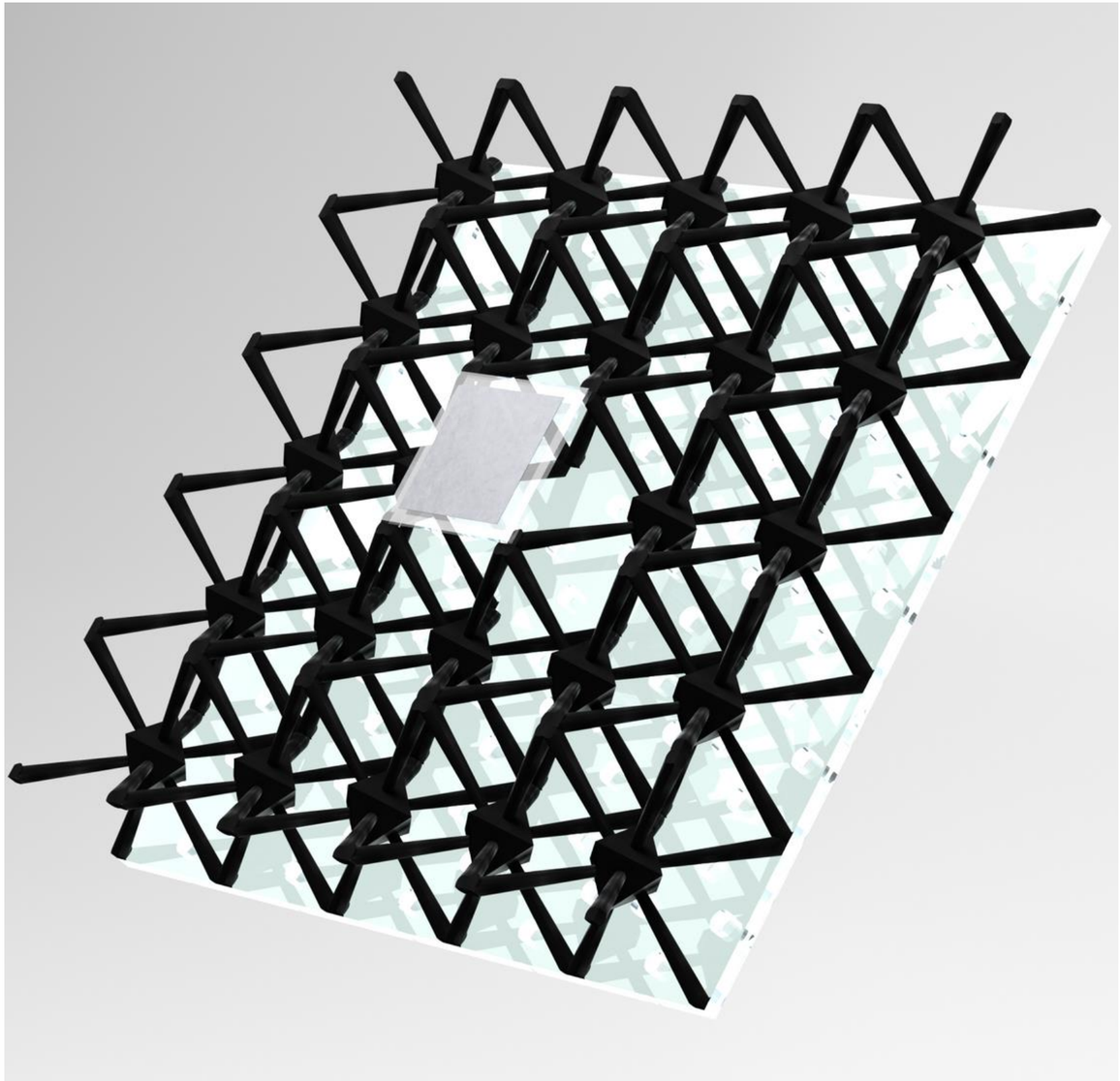
FINAL DESIGN

Each Module is consist of 15cm x 15cm TPU injection molded phonic crystal structure, 10mm pvdf film, and acrylic board.



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IP AND EXPLOITATION

META-MATERIAL MARKET

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32.01%

Annual growth rate (CAGR)

0.9B TO 14.5B

in 2022 in 2032

KEY MARKET PLAYERS

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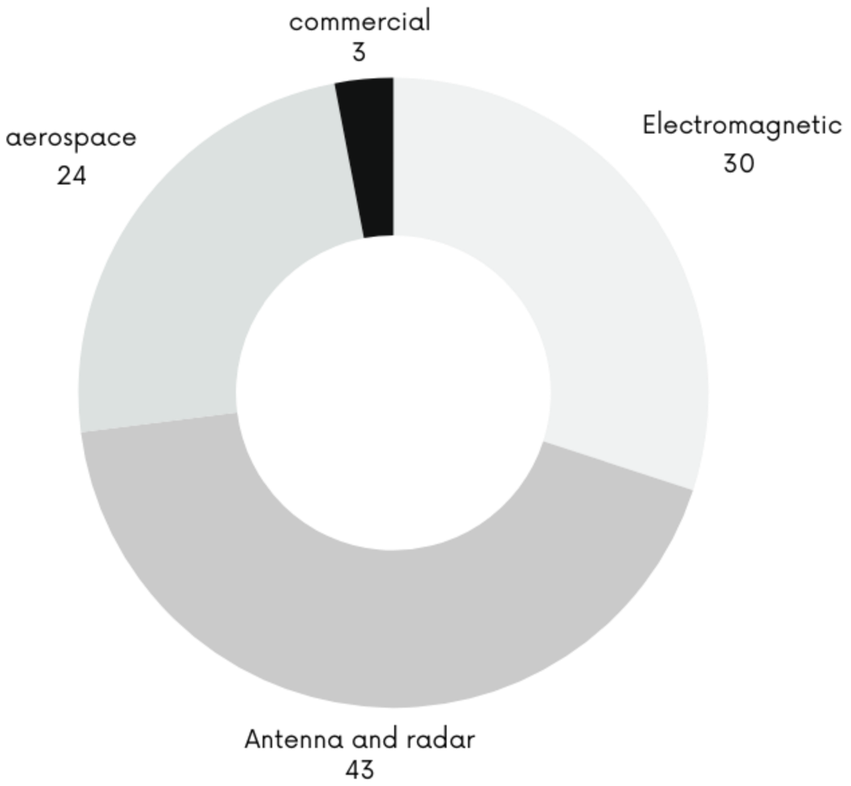
AMG MERFORD



Sunrising, unsaturated

Mature, Developed

ACOUSTIC INSULATION

glass wool, synthetic fibers, chemicals...
all from crude oil

We use **recycled TPU and TPE, eco-friendly**

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VISION

B2B

Industries with high noise levels and potential for energy harvesting.

MISSION

SAFETY

Addressing the overlooked noise pollution issue

APPROACH

SUSTAINABLE

Circular Economy, Energy reproduction, IoT

BENCHMARKING

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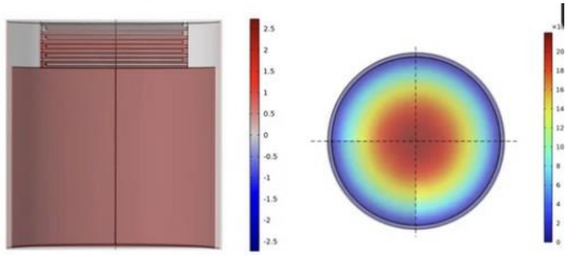
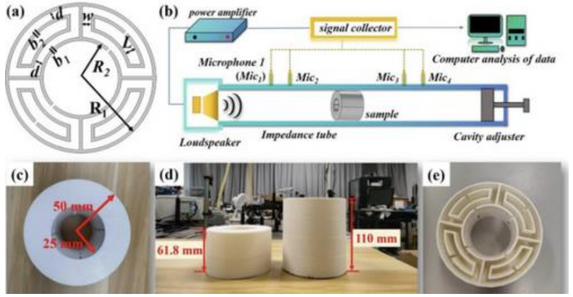
DISSIPATE

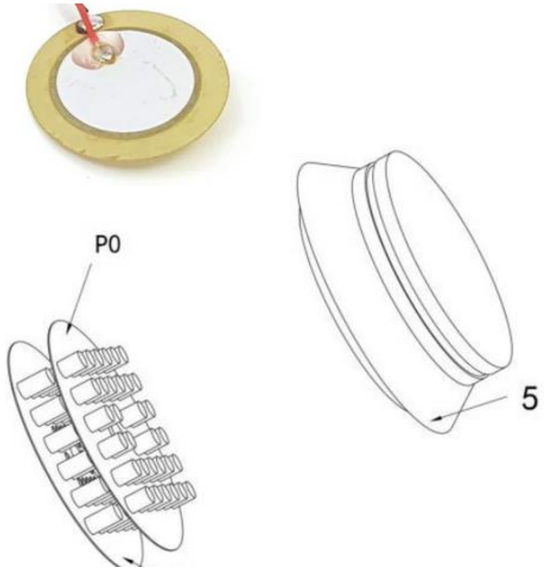
space coiling , porous material





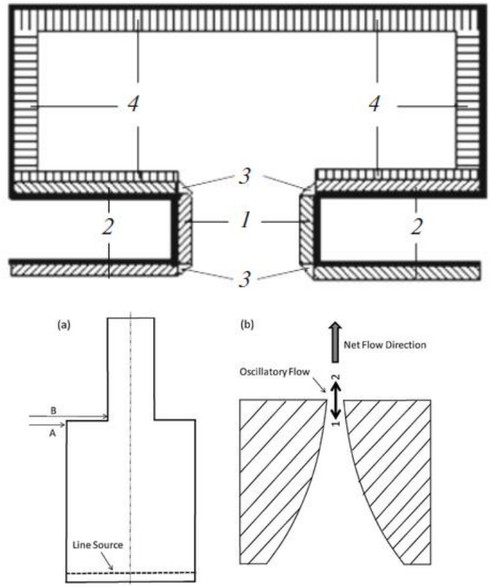
INEFFICIENT

Ceramic piezo disc



HIGH LOSS

Helmholtz resonator



COMPANY EXPLOITATIONS

CONNECT

Net Zero Initiative Programme

PARTNER

Field Testing and initial marketing

SUPPLY

Provider of film and price negotiation

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WHERE WILL WE BE IN 5 YEARS?

Envision, explore, expand

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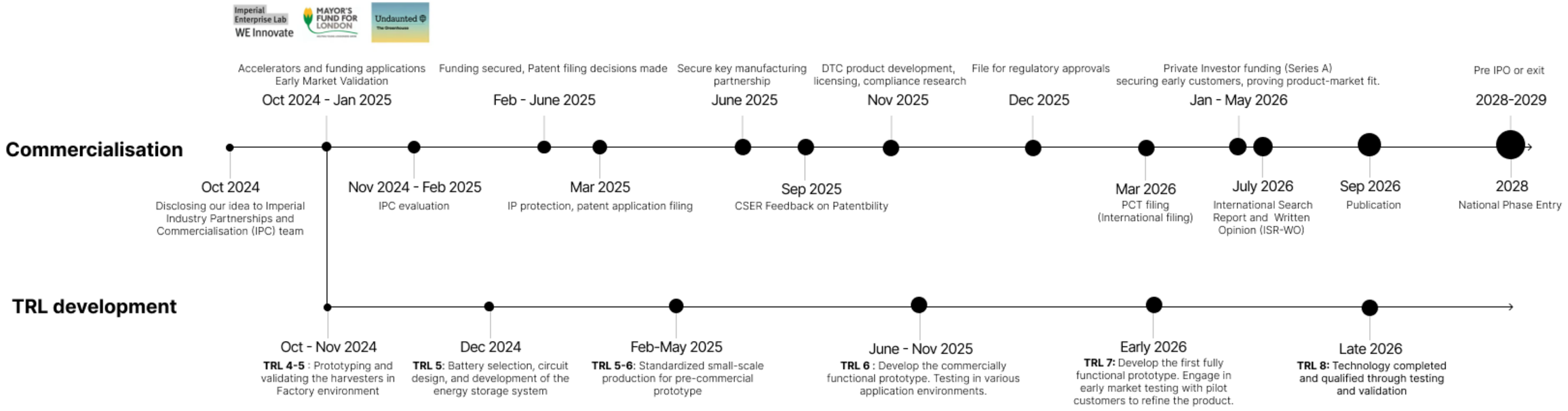
THE TIMELINE